

ADITYA RAJ KAR

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Summary

Detail-oriented Software Developer with hands-on experience building scalable web applications, real-time IoT solutions, and data-driven systems. Proficient across the stack with modern languages and frameworks (C++, Python, React). Passionate about system architecture and leveraging advanced technologies to solve complex architectural challenges.

Education

VIT, Vellore – B.Tech CSE (Internet of Things) **2023 – 2027**

CGPA: 8.44/10

Amity International School, New Delhi – XII CBSE **2021 – 2023**

Score: 88.6%

Narayana E-Techno School, Gurugram – X CBSE **2021**

Score: 93.2%

Experience

Software Intern – Perennial Systems Pvt. Ltd.

June 2025 – July 2025

Hybrid

- Worked with **ESP32/ESP8266/Arduino** for IoT prototyping, sensor integration, and automation use cases.
- Developed cloud dashboards for real-time data logging, visualization, and alerts.
- Documented processes to ensure project clarity and knowledge sharing.

Projects

OpinionMap **React, TypeScript, FastAPI, LangGraph, Gemini, ChromaDB, Docker**

- Engineered a containerized full-stack web application using React and FastAPI, integrating Prometheus and Grafana for real-time system monitoring and observability.
- Architected a multi-agent AI workflow using LangGraph and Google Gemini to autonomously collect, process, and analyze unstructured data from diverse social media sources.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline with ChromaDB, leveraging vector similarity search to ground AI-generated market intelligence reports in factual source data.

BUSIT — Digital Shuttle Credit System **Next.js 16, TypeScript, PostgreSQL, Prisma, Tailwind CSS**

- Engineered a custom digital wallet platform to digitize campus transit, building an in-house browser-based camera scanner and dynamic QR generation system to eliminate hardware dependencies.
- Architected a secure PostgreSQL database with bcrypt-based credential hashing and developed an automated SMTP email pipeline for real-time transaction receipts.
- Developed the core transaction engine, enabling real-time verification of student boarding passes and instant digital wallet fare deductions upon scanning.

Smart Room Dashboard – IoT Automation **ESP32, Firebase, HTML, JavaScript**

- Created a smart room simulation using ESP32 and Firebase to automate fan and light control based on motion and temperature data, with real-time dashboard toggles and automation overrides.

Technical Skills

Languages: C, C++, Python, TypeScript, JavaScript, SQL
Frameworks: React, Next.js, FastAPI, Tailwind CSS, Prisma, LangGraph
Databases: PostgreSQL, ChromaDB, Firebase
Tools: Git, GitHub, Docker, Prometheus, Grafana, Vercel, Netlify

Certifications

- Placement Preparation using DSA with C++ by DevTown
- OCI Foundations Associate by Oracle
- MERN Full Stack by Ethnus
- Young Professional Career Development Course by TCS iON